

A Network Analysis of the Western US Power Grid

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Abstract

Network theory is a high performance tool for analyzing and interpreting structural properties of complex systems. Understanding large technological networks, such as the Western US power grid (WUSPG), can help understand the fundamental properties of modern infrastructure. In this report, we analyze the structure of the WUSPG using complex network theory tools to give quantitative results of the system's robustness, to then be contrasted with adequate null models to check their statistical significance. The results reflect how power grids were constructed to reinforce system stability and minimize failure propagation, illustrating how complex systems theory can be applied to improve and understand critical infrastructure.

I. INTRODUCTION [1][2]

Modern society depends on critical infrastructures, particularly electrical power grids, which function as large-scale technological networks that deliver the necessary energy to economic activity, public health, and communication. The structure of these networks is not arbitrary, it is the result from engineered constraints and historical development. They present features such as small-world properties, hierarchical organization, and degree heterogeneity that distinguish them from random graph models.

This study applies complex network theory to analyze the Western US power grid (WUSPG). By analyzing the topological properties of the network, such as degree distributions, clustering, centrality, and community structure, we identify how technological networks self-organize under operational constraints. Specifically, we address: (1) How do degree distributions deviate from null models such as Erdős-Rényi or scale-free networks? (2) How is centrality distributed across the network? (3) What mesoscale communities exist and how do they relate to the global structure?

We can interpret the core structural descriptors of the WUSPG through complex network theory to reveal and understand universal organizing principles and distinctive topological features of technological networks.

II. METHODS

This section outlines the methodological approach of our study. We begin reviewing key concepts from network theory to establish foundational principles and community detection algorithms. Then we follow by describing the computational methods used as well as the dataset chosen for the analysis.

We present the following notation that will be used throughout the text: $\#$ nodes $\equiv n$ and $\#$ edges $\equiv m$

A. Fundamental definitions in Network Theory

Formal definitions of networks [3]

Def: 1 (Network). A **network** G is a pair (V, E) . $V \equiv$ vertex set; with the elements being the **nodes** of the network. $E = V \otimes V \equiv$ edge set; with the elements being the **edges** of the network.

Def: 2 (Undirected Graph). A network $G = (V, E)$ whose E is symmetric.

Def: 3 (Proper Edge). An edge that joins two distinct vertices.

Def: 4 (Self-Loop). An edge that joins a single end-point to itself.

Def: 5 (Simple Graph). A network G with no self-loops or multi-edges.

General descriptors of a network [1][3][4]

Def: 6 (Adjacency Matrix). Suppose $G = (V, E)$ is a simple network where $V = \{1, 2, \dots, n\}$. For $1 \leq i, j \leq n$:

$$a_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases} \quad (\text{II.1})$$

Then the square matrix $A = (A_{ij})$ is called the **Adjacency Matrix** of G .

Def: 7 (Density). The fraction of those edges that are actually present compared to the maximum possible value in an undirected simple network:

$$\rho = 2m/n(n-1)$$

Def: 8 (Degree). In a simple undirected graph, the **degree** of a vertex v , k_v , is the number of edges incident on v .

Clustering Metrics [1]

Def: 9 (Transitivity). It is the property that if node u is connected to v and v is connected to w , then u is also connected to w .

Def: 10 (Local Clustering coefficient). Denoted C_i , it is defined as the fraction of pairs of neighbours of node i that are actually connected to each other.

$$C_i = \frac{2m_i}{k_i(k_i - 1)}$$

Def: 11 (Global Clustering coefficient). Defined as the average local clustering coefficient

$$C = \frac{1}{n} \sum_{i=1}^n C_i$$

Distance Metrics [3][4]

Def: 12 (Walk). A series of edges (not necessarily distinct): $(u_1, v_1), \dots, (u_p, v_p)$, such that $v_i = u_{i+1}$ for $i = 1, \dots, p-1$, $\forall u_i, v_i \in V$.

Def: 13 (Length). $\#$ p of edge-steps in a walk.

Def: 14 (Trail). A walk without repeated edges.

Def: 15 (Path). A trail with all distinct u_i .

Def: 16 (Distance). The **distance** $d(u, v)$ between two vertices u, v in a network G is the length of the shortest walk between u and v .

Def: 17 (Eccentricity). Denoted $\text{ecc}(v)$, is the maximum distance from v to any other vertex x in the graph.

$$\text{ecc}(v) = \max_{x \in V} \{d(v, x)\}$$

Def: 18 (Diameter). Denoted $\text{diam}(G)$, is the maximum eccentricity among all vertices in G , representing the largest distance between any pair of nodes.

$$\text{diam}(G) = \max_{v \in V} \{\text{ecc}(v)\} = \max_{x, y \in V} \{d(x, y)\}$$

Def: 19 (Radius). Denoted $\text{rad}(G)$, is the minimum eccentricity among all vertices in G .

$$\text{rad}(G) = \min_{v \in V} \{\text{ecc}(v)\}$$

Centrality Measures [1][3]

Def: 20 (Degree centrality). The number of direct connections a node has, its degree.

Def: 21 (Closeness centrality). How quickly a node can reach all other nodes via shortest paths.

Def: 22 (Betweenness centrality). The extent to which a node lies on shortest paths between other node pairs.

Def: 23 (Eigenvector centrality). A node's importance based on the importance of its neighbors.

Def: 24 (Katz centrality). Eigenvector centrality with an added constant term so all nodes have non-zero scores.

Def: 25 (PageRank centrality). Katz centrality normalized by a node's out-degree to prevent high-degree nodes from dominating.

Def: 26 (Subgraph centrality). A node's participation in all closed walks within the network.

Partition descriptors [1][5]

Def: 27 (Modularity). Denoted Q , quantifies the level of non-randomness in the placement of edges in the network

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta_{g_i g_j}$$

Def: 28 (Internal edge density). Denoted $\delta_{int}(\mathcal{C})$, is the ratio between the number of internal edges of subgraph $\mathcal{C}$ and the number of possible internal edges

$$\delta_{int}(\mathcal{C}) = \frac{\# \text{ internal edges of } \mathcal{C}}{n_c(n_c - 1)/2}.$$

Def: 29 (Conductance). Denoted $\phi(S)$, quantifies the quality of a community S , by computing the ratio of edges leaving a community to its total internal connectivity

$$\phi(S) = \frac{c_S}{2m_S + c_S}$$

where c_S is the number of internal cut edges and $2m_S$ the number of internal degrees (twice the internal edges).

B. Community detection methods [1][6][7]

Def: 30 (Louvain Algorithm). An agglomerative algorithm which joins single nodes into groups, then groups into groups and so on, by maximizing modularity.

Def: 31 (Greedy Modularity Optimization). Another bottom-up clustering method that forms communities by successively merging node groups to optimize modularity.

Def: 32 (Label Propagation). Every node is initially assigned a unique label. In iterative steps, each node updates its label to the most common among its neighbours until convergence where no label change occur.

Def: 33 (Asynchronous Label Propagation). Variation of LP, where node labels are updated sequentially one by one in a random order. As soon as a node updates its label, that new label is visible to its neighbours for their own updates in the same iteration.

Def: 34 (Fluid communities). A fixed number of communities are modeled as “fluids” expanding in a container (the graph). Fluids compete for vertices based on density, denser fluids attract vertices more strongly until boundaries stabilize.

C. Null Models [3]

Def: 35 (Erdős–Rényi Network Model). A random graph model where edges are formed between every possible pair of n nodes with a fixed, independent probability p .

Def: 36 (Barabási–Albert Network Model). A generative model that produces scale-free networks through growth (adding new nodes one by one) and preferential attachment (new nodes connect to existing nodes with probability proportional to their current degree).

D. Data Collection and Processing [8]

The data of the WUSPG used in this work was provided by the Netzschleuder network repository. The data is a collection of nodes, which are transformers or power relay points; and edges, where two nodes are connected if a power line runs between them. This network is undirected, which means that the nodes are connected to each other in both directions.

The analysis was performed using *Python* as the programming language and *NetworkX* as the main package to calculate all the parameters and visualizations shown in this report. The code that runs all results can be consulted in the other attached file *individualProject.ipynb*.

III. BASIC DESCRIPTORS

The WUSPG is a network with $n = 4941$ nodes and $m = 6594$ edges, where no nodes are left isolated. This directly means that the giant component is equal to the total network, and so all these data will be used in the following analysis. The network exhibits an extremely low density ($\rho = 0.000540$) and it’s said to be sparse and the fraction of non-zero elements in the adjacency matrix tends to zero.

Figure 1 provides a visualization of the whole network using the *spring* layout, where the node size represents the degree of the nodes and node colour represents the node centrality. Both of these parameters will be studied in deeper detail in future sections.

IV. CLUSTERING

The study began with a multi-level clustering analysis, evaluating transitivity and the distribution of local clustering including the mean, median, and standard deviation of the local coefficients.

The transitivity or global clustering is a measurement that compares the number of closed triangles in the network with the maximum potential number of triangles if the network were fully connected. The local clustering measures how connected is node to its neighbours. The average clustering coefficient will be simply the geometric mean of the values, which is equal to the mean local clustering.

Western US Power Grid

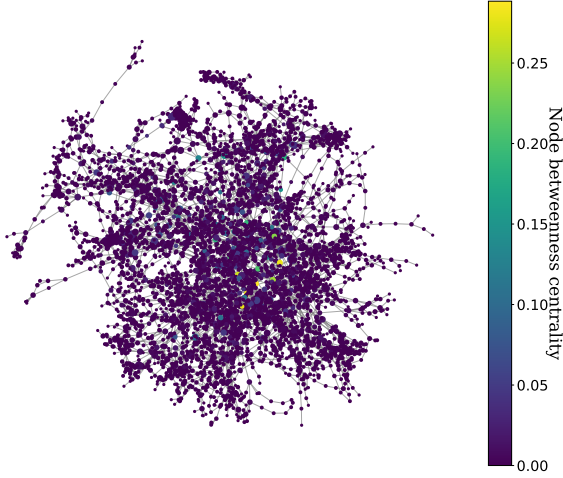


Figure 1: Western US power grid represented using NetworkX

Metric	Value
Transitivity	0.1032
Global Clustering Coefficient	0.0801
Median Local Clustering	0.0000
Std. Dev. Local Clustering	0.2242

Table 1: Summary of Network Clustering Statistics

In figure 1, the shaded area represents the clustering coefficients that fall in the interval $[\mu - \sigma, \mu + \sigma]$. **Note:** We only show the interval $[0, \mu + \sigma]$, as negative clustering coefficients are meaningless. We check that the clustering is very low sitting very close to zero. There are approximately 4000 local coefficients within 0 and 0.02 and the median falls at exactly 0 reinforcing that the network shows very low clustering.

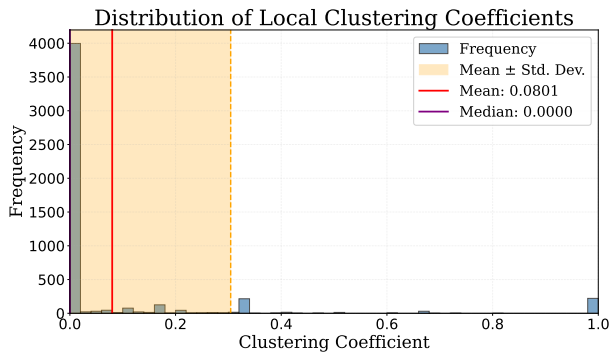


Figure 2: Local clustering distribution

V. DISTANCES

In this section we outline the basic distance parameters including the average shortest-path length, diameter, radius and average eccentricity.

Metric	Value
Average Shortest Path Length	18.9892
Network Diameter	46
Network Radius	23
Average Eccentricity	34.5408

Table 2: Summary of Network Distance Metrics

To plot the a histogram of the shortest path length distribution we have sampled the network to reduce computation time. We randomly and uniformly choose $n = 1000$ nodes and measure the shortest path lengths for all sample nodes.

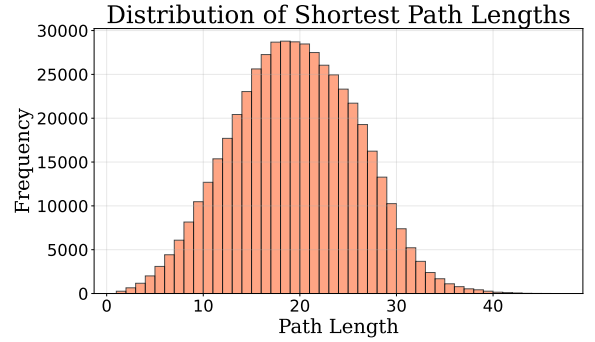


Figure 3: Sample path length distribution for $n = 1000$ nodes belonging to the network.

The distribution exhibits a bell-shape, indicating the network has a cohesive structure. This means that most nodes are available to another within a comparable number of steps. The distribution presents a maximum at 19 which strongly agrees with the average shortest path length 18.99. This central tendency suggests that the network is relatively homogeneous, lacking significant fragmentation that would otherwise be perceived as a bimodal distribution.

Even though the maximum number of steps between two nodes (diameter) is 46, most connections fall between the range of 10–30 steps. The positive skew (long tail) represents that extreme distances are mostly outliers rather than a structural feature of the system. This implies that despite its size, the network maintains efficient connectivity, with very few nodes being peripherally isolated from the core.

VI. DEGREE STATISTICS

The table below holds the results of the average degree, median degree, standard deviation, as well as the maximum and minimum degrees of the nodes within the power grid network.

Metric	Value
Average Degree ($\langle k \rangle$)	2.6691
Median Degree	2.0000
Standard Deviation (σ_k)	1.7913
Minimum Degree (k_{min})	1
Maximum Degree (k_{max})	19

Table 3: Summary of Network Degree Statistics.

The degree statistics present a picture of a sparse and heterogeneous network where nodes have, on average, fewer than three connections ($\langle k \rangle \approx 2.67$). The fact that the median degree (2.00) is lower than the average indicates a right-skewed distribution, suggesting the topology is dominated by many weakly connected nodes alongside a minority of more highly connected “hubs” that reach a maximum degree of 19. Additionally, the standard deviation is 1.79, relatively high compared to the mean, confirming that connectivity is unevenly distributed, rather than being uniform across the graph.

We have chosen multiple possible fits for the Probability Density Function (PDF) and Complementary Cumulative Distribution Function (CCDF). These include:

Power-law: $P(k) = Ck^{-\alpha}$

Exponential: $P(k) = C \exp(-\lambda k)$

Log-normal:

$$P(k) = \frac{A}{k\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln k - \mu)^2}{2\sigma^2}\right)$$

and for the CCDF:

$$F(k) = A \left(1 - \Phi\left(\frac{\ln k - \mu}{\sigma}\right)\right)$$

where Φ denotes the standard normal cumulative distribution function.

Based on the degree distribution plots and fitting statistics, the network’s connectivity structure is best described by a Log-normal or Exponential distribution rather than a scale-free Power-law model.

The PDF histogram reveals a right-skewed degree distribution with a prominent peak at low degrees

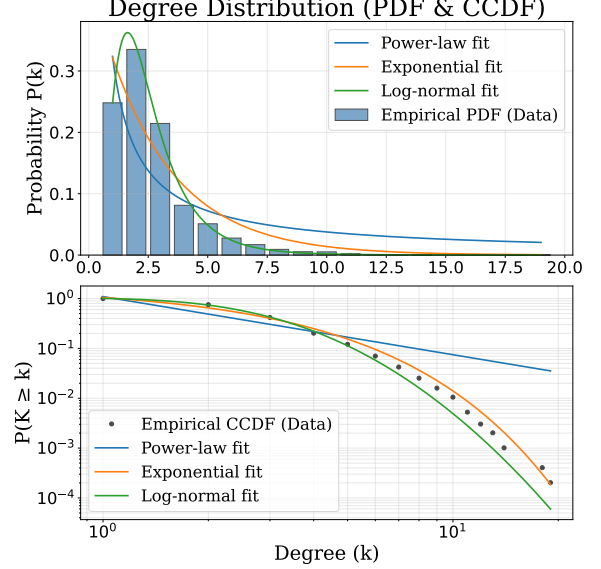


Figure 4: Multiple fits for the PDF and CCDF.

Model	Fitted Equation / Parameters	R^2
<i>Probability Density Function (PDF)</i>		
Power-law	$P(k) = 0.3237 \cdot k^{-0.9344}$	0.6790
Exponential	$P(k) = 0.4564 \cdot e^{-0.3490k}$	0.8571
Log-normal	$A = 0.97, \mu = 0.80, \sigma = 0.56$	0.9961
<i>Complementary Cumulative (CCDF)</i>		
Power-law	$F(k) = 1.0934 \cdot k^{-1.1655}$	0.0539
Exponential	$F(k) = 1.6949 \cdot e^{-0.4796k}$	0.9801
Log-normal	$A = 1.03, \mu = 0.98, \sigma = 0.51$	0.9372

Table 4: Quality of fit statistics for degree distribution models

and a long tail. The Log-normal model provides the best fit ($R^2 = 0.9961$), accurately capturing the distribution’s curvature, while the Power-law model ($R^2 = 0.6790$) significantly underperforms. This result indicates that the network’s degree heterogeneity stems from multiplicative growth processes rather than preferential attachment or random mechanisms.

This conclusion is reinforced by the CCDF analysis, where the Power-law fit fails almost completely ($R^2 = 0.0539$), indicating that the probability of finding highly connected hubs drops off much faster than a scale-free network would predict. In this cumulative view, the Exponential model actually performs best ($R^2 = 0.9801$), suggesting that while the network is heterogeneous, the maximum degree is strongly bounded and “super-hubs” are statistically unlikely.

VII. CENTRALITY STRUCTURE

The centrality structure of the US power grid network was determined for various different measurements: degree (D), closeness (C), betweenness (B), eigenvector (E), Katz (K), PageRank (P) and subgraph (S) centrality. The centralities were determined using the built-in functions of the *NetworkX* library. We rank the top 25 nodes for each centrality measure in table 6 in *Appendix B*.

We have also produced an overlap matrix that counts the number of nodes that appear in both rankings for pairs of centrality measures.

Overlap Matrix of Top 25 Nodes in Centrality Measures

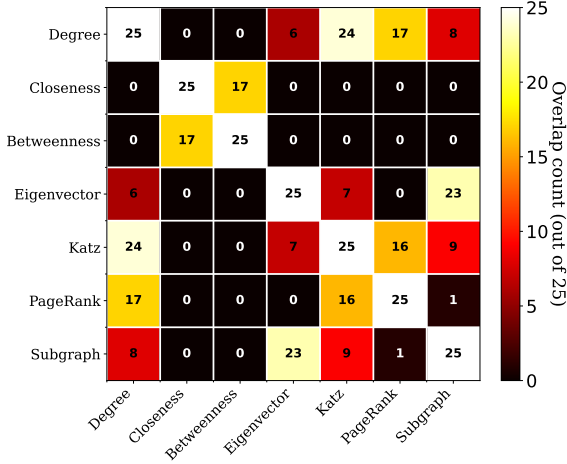


Figure 5: Overlap matrix of the top 25 nodes for different centrality measures.

The overlap matrix reveals patterns in centrality measures. There’s a great agreement between degree and Katz (24 shared nodes), and eigenvector and subgraph centrality (23 shared nodes), indicating these pairs capture similar node importance aspects. In contrast, zero overlap between eigenvector centrality and closeness, betweenness, or PageRank signals these metrics capture orthogonal dimensions of network importance. Betweenness and closeness centrality stand alone, showing minimal overlap with all other centrality measures (0 nodes) except themselves, demonstrating that nodes with high betweenness/closeness, (those lying on many shortest paths) are not highly connected or locally embedded. The matrix illustrates that no single centrality metric fully characterizes nodal roles and so a multi-dimensional analysis might be useful to fully understand node importance.

The top twenty-five rankings table (see appendices) reveals substantial heterogeneity across centrality

measures, with node 2554 dominating degree, betweenness, and PageRank centrality, while nodes 4459, 832, and 4346 occupy leading positions in multiple metrics. Many nodes appear in only one or two rankings, indicating that different centrality concepts capture distinct aspects of nodal importance.

VIII. COMMUNITIES

There are many methods that can be implemented to study community structure. A priori, we do not know which one best adapts to the WUSPG so we have tried the 5 following methods: the Louvain Algorithm (LA), the Greedy Modularity (GM) Optimization, Label Propagation (LP), Asynchronous Label Propagation (ALP) and Fluid Communities (FC).

The Louvain algorithm has been implemented from a python library named *python-louvain*, and the other four are directly imported from the *NetworkX* library.

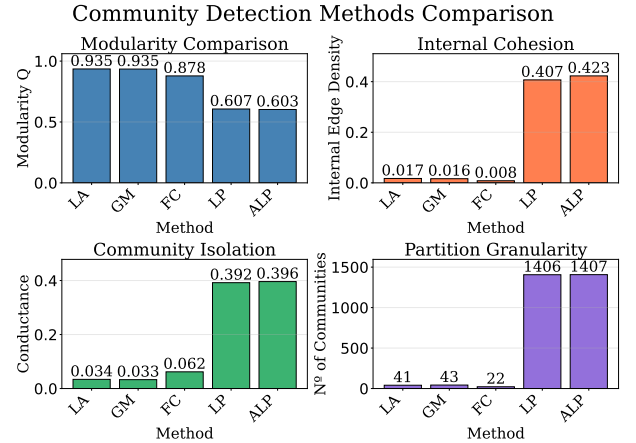


Figure 6: Graphical comparison of the different values obtained for the community detection algorithms.

By comparing their results to choose the best one, we find that the Louvain Algorithm yields the greatest modularity value ($Q = 0.9354$), indicating the strongest large-scale community structure among all tested methods. Greedy modularity maximization achieves a very similar modularity score, but Fluid communities and the two label-propagation variants drop to lower Q values, which means the partitions they detect, are weaker modular organizations.

Observing the internal density plot, we have that LP and ALP create very dense communities ($\delta_{int}(C) \approx 0.4$), whereas Louvain, Greedy Modularity and Fluid produce much sparser modules with densities below 0.02. However, these high densities come at the cost of increasing the number of communities (~ 1400)

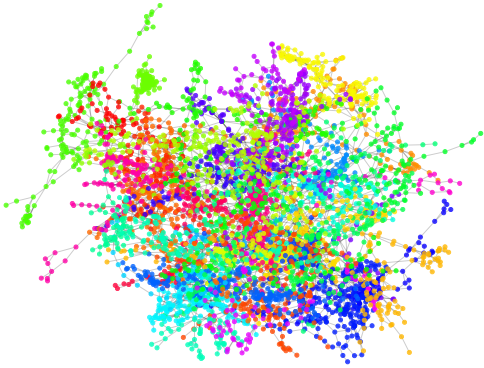


Figure 7: Representation of the different communities within the WUSPG

but they yield very small groups with only a few nodes. In the other hand, Louvain, Greedy and Fluid keep a lower number of communities (~ 20 – 40) of sizeable extent.

Regarding conductance, Louvain and Greedy Modularity have the lowest values ($\phi(S) \approx 0.3$). As it measures how well the communities are separated from the rest of the network, this directly indicates that the partitions found by these algorithms best represent the actual modules within the network. For Fluid this value is a bit higher, and much higher for label-propagation methods (≈ 0.39), reflecting that their clusters are poorly isolated and strongly inter-linked with neighboring nodes.

Overall, the method which achieves the best results is the Louvain algorithm. The visualization of its partitions can be found in figure 7 with different colours to label each community. These coloured “clouds” are internally tangled and well mixed, but are still visually separable from one another. They form overlapping lobes and branches that cover the whole graph without leaving isolated islands.

The embedding also reveals a sparse, elongated infrastructure with numerous peripheral chains attached to a denser backbone. This is consistent with a grid where long transmission lines connect regional clusters of substations and generators.

This method splits the network into communities of various different sizes ranging from 18 to 245 nodes. The sizes are still comparable and avoid the partition of the network into singletons or groups of very few nodes, that may not actually be enough to represent measurable structure within the power grid.

IX. NULL MODELS

The above findings must be contrasted with appropriate null models to identify the statistical significance of the results, or whether it was a matter of random chance. The most typical null models are the Erdős–Rényi (ER) random networks and Barabási–Albert (BA) preferential attachment networks. We generated 10 of each with same size (n nodes and m links) as the WUSPG. The BA networks were generated with $m_0 = 2$, to add two edges on each step, so that clustering within the networks was possible. Metrics computed on these null models were averaged across the ensemble to provide a baseline for benchmarking the real network’s properties.

Table 5 collects the metrics that characterize the network’s topology as follows: average degree $\langle k \rangle$, network density ρ , average clustering coefficient $\bar{C}$, average shortest path length $\bar{L}$, network diameter D and the modularity Q . The data for the mean and standard deviation of the 10 realizations of each method is shown.

Metric	Real Data	ER Random Mean (SD)	BA Scale-Free Mean (SD)
$\langle k \rangle$	2.6691	2.6578(0.0328)	3.9984(0)
ρ [10^{-4}]	5.4030	5.3801(0.0660)	8.0912(0)
$\bar{C}$ [10^{-2}]	8.0104	0.0494(0.0274)	0.8125(0.0957)
$\bar{L}$	18.9892	8.5075(0.1307)	4.7666(0.0369)
D	46.0000	20.0000(0.9428)	8.4000(0.5164)
Q	0.9371	0.7019(0.0076)	0.5356(0.0014)

Table 5: Comparison of topological metrics: real vs. null model networks.

We have broken down the discussion into subsections to emphasize focus in each of the parameters.

A. Basic structure

The real WUSPG has an average degree of $\langle k \rangle \approx 2.67$ and a very low density of $\rho \approx 5.4 \times 10^{-4}$. The ER ensemble has very similar results with $\langle k \rangle \approx 2.66$, with a density of $\rho = 5.4 \times 10^{-4}$. Whereas BA networks yield a higher $\langle k \rangle \approx 4.00$, and density $\rho = 8.1 \times 10^{-4}$. This means that the BA model does not match the empirical edge count, which is expected as we chose $m_0 = 2$, so the generated networks were denser than the real power grid.

B. Clustering and local cohesion

The average clustering coefficient of the real grid is $\bar{C} \approx 0.0801$, which is an order of magnitude larger than that of the null models. ER networks exhibit $\bar{C} \approx 4.9 \times 10^{-4}$, while BA networks have

$\bar{C} \approx 8.1 \times 10^{-3}$, both with small standard deviations, indicating that such low clustering is intrinsic to those models. This indicates that the real power grid has more cohesive local neighborhoods than random or scale-free preferential attachment can generate.

C. Distances and diameter

The mean shortest-path length of the real network is $\bar{L} \approx 18.99$ with a diameter of $D = 46$, demonstrating a globally stretched topology. In contrast, the ER ensembles yield $\bar{L} \approx 8.5$ with diameters around $D \approx 20$, while BA networks are even more compact, with $\bar{L} \approx 4.77$ and $D \approx 8.4$, characteristic of strong hub-mediated shortcuts. The small standard deviations for both null models compared with their large gap to the empirical values show that the real grid is systematically less efficient in terms of path length than graphs generated by ER or BA models.

D. Degree Distribution

The figures illustrating the PDFs and CCDFs, together with their corresponding fits for the ER and BA ensembles, are presented in *Appendix B*. The degree distribution fittings provide evidence for spatial constraints suppressing a scale-free topology. The real network’s empirical data rejects the power-law model in the CCDF ($R_{\text{CCDF,PL}}^2 \approx 0.054$) but is very nicely captured by the log-normal distribution ($R_{\text{PDF,LN}}^2 \approx 0.996$ and $R_{\text{CCDF,LN}}^2 \approx 0.937$), and by the exponential model ($R_{\text{CCDF,Ex}}^2 \approx 0.980$).

ER random networks present acceptable fit statistics for a power-law ($R_{\text{PDF,PL}}^2 = 0.613$) but excel at log-normal fits ($R_{\text{PDF,LN}}^2 = 0.985$), reflecting their near-uniform random assembly. The BA model, by contrast, produces a perfect power-law fit ($R_{\text{PDF,PL}}^2 = 1.000$) and excellent log-normal ($R_{\text{PDF,LN}}^2 = 1.000$) across the PDF. The formation of hubs predicted by preferential attachment produces these results, yet the real grid suppresses them through spatial distance costs and physical capacity limits. This is shown when observing the CCDF power-law fit ($R_{\text{CCDF,PL}}^2 = 0.766$), as it significantly outperforms the real network’s ($R_{\text{CCDF,PL}}^2 = 0.054$). The BA’s exponential shows negative fit quality ($R_{\text{CCDF,Ex}}^2 \approx -0.818$) while the real network’s is ($R_{\text{CCDF,Ex}}^2 \approx 0.980$). This indicates rapid degree saturation absent in BA’s scale-free tail, confirming that infrastructure constraints.

E. Centrality Structure

The figures holding the mean centrality matrices for the ER and BA ensembles can be found in *Appendix B*. The matrices reveal notable differences in nodal

importance between the real power grid and null models. The real network exhibits highly selective and specialized centrality roles, where some centrality measures show a very strong agreement while others share minimal overlap. This indicates that different important nodes have functionally distinct roles. There exist nodes that are highly connected hubs, some are critical for path routing, and others provide local structural reinforcement.

In contrast, both the ER and BA null model display much more uniform overlap patterns across all centrality pairs, with overlap counts (4–24 nodes per pair). The ER model lacks cohesive centrality structure (4–20 nodes shared per metric pair), reflecting randomness in node selection without functional specialization. The BA model exhibits high agreement within degree-like metrics (24 nodes shared between degree, Katz and PageRank) but more variable overlap elsewhere (7–18 nodes), reflecting its hub-and-spoke structure but lacking the orthogonal dimensions seen in the real grid. This demonstrates that the real power grid’s spatial and engineering constraints create distinct centrality hierarchies unavailable to random models.

F. Community structure and modularity

The modularity of the real grid is very high, with $Q \approx 0.94$ yielded by the Louvain algorithm. This indicates a pronounced community structure. In contrast, ER networks show weaker modularity, $Q \approx 0.70$, where communities mostly reflect random fluctuations rather than strong structural organization. BA networks show an even lower modularity, $Q \approx 0.54$. This is mainly due to the hub-and-spoke architecture, and not to geographical or functional regions as in the actual power grid.

X. DISCUSSION

A. Topology and Degree Distribution [9][10][11][12]

The WUSPG is shaped by physical and spatial constraints that prevent the emergence of scale-free structure. The poor power-law fit to the degree CCDF and excellent exponential and log-normal fits show that “super-hubs” are strongly suppressed. This is consistent with spatially embedded infrastructure networks, where link formation is limited by geographic distance, costs, and transmission losses. This favours short-range connections and limits the maximum degree at a moderate value, explaining why the real grid is far from the heavy-tailed, hub-dominated regime.

B. Deviation from ER and BA Null Models [6][11][12]

When contrasted with ER and BA ensembles, the real grid shows systematically different higher-order structure. Clustering is about an order of magnitude larger in the real network than in ER and BA graphs, reflecting the tendency of geographically nearby substations to form tightly knit local triangles rather than random or purely degree-driven links. At the same time, the real grid has much longer average path length and diameter than either of the null models, whose small-world shortcuts arise from unrestricted long-range connections. Modularity is also particularly high, indicating well-defined communities that align with operational zones, whereas ER graphs show weak modularity and BA graphs produce communities dominated by hubs rather than geographically meaningful regions. Together, these comparisons show that random wiring and preferential attachment miss key features of real transmission grids.

C. Generative Mechanisms, Limitations, and Outlook [10][11][12][13]

The observed combination of log-normal degree statistics, high clustering, long path lengths, and strong modularity suggests that power grid topology arises from spatially constrained, continuous growth rather than from a single universal mechanism. The current analysis is limited by the lack of node-level metadata (e.g. coordinates) and by its focus on unweighted, static structure, which prevents direct validation of how communities map to real control areas or how topology interacts with power flows. Future work should integrate geographic and electrical data to study dynamic evolution and link topological patterns with robustness, congestion, and cascading failure risk.

XI. CONCLUSION

This study reveals that the WUSPG characteristic features shaped by engineered constraints rather than random processes or preferential attachment. The network suppresses the formation of super-hubs, exhibits pronounced clustering and contains highly modular communities. Comparison with ER and BA null models demonstrates that neither random graphs or preferential attachment mechanisms capture the empirical topology. These results establish that technological infrastructure networks must satisfy objectives such as geographic proximity, capacity limits, and cost optimization, much more complex than simple connectivity rules.

However, this analysis is limited by the absence of geographical coordinates and exact node labeling (generator, substation, relay point), which prevents direct validation of network importance against real infrastructure data. Future work should incorporate spatial coordinates to contrast topological centrality with geographic locations, integrate edge weights representing transmission line capacity, and extend the analysis to temporal networks. The design, maintenance, and operation of power grids can be strongly supported by complex systems theory, linking topological principles to real-world engineering challenges.

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Appendix A: Data Processing [8]

As stated before, the data was collected from the Netzschleuder network repository. The code is **original** and can be found in the attached file *PowerGridAnalysis.ipynb*. Many measures are implemented using the built-in *Python* functions from *NetworkX* and other libraries.

The visualizations and plots have also been produced using *Python* libraries such as *Matplotlib* and *NetworkX*. For each section within the code, the functions produce *.csv* (for storing numerical results) and *.pdf* for the visualizations. See the attached file *README.md* for further details on how to reproduce the results found in this report.

Appendix B: Supplementary Information

This appendix holds secondary figures that are relevant to the analysis discussed in this report but, to keep the extension shorter, have been stored within the following section.

Centrality Structure

The table for the Top 25 nodes ranked by the different centrality measures is found below. The node labels are the labels given by default when loading the network.

Rank	D	C	B	E	K	P	S
1	2554	1309	4165	4382	2554	4459	4346
2	4459	2595	2544	4346	4459	832	4382
3	832	2606	1244	4337	4346	3469	4353
4	3469	1132	4220	4333	3469	2554	4385
5	4346	2607	2529	4353	832	1225	4337
6	2543	1244	1268	4385	2543	598	4396
7	2383	1477	1309	4403	2586	2383	4333
8	2576	2558	1245	4348	2576	2576	4374
9	2586	2529	427	4396	3896	2440	4403
10	3896	2533	2607	4374	2383	3896	4362
11	1225	2613	2595	4362	2663	3356	4402
12	2618	2544	2606	4399	2435	2435	4348
13	2435	4220	70	4402	2618	2543	4393
14	2663	1126	109	4414	2440	2618	4414
15	2440	2619	1168	4409	1225	2648	2554
16	2283	4165	4121	4336	4382	2283	4409
17	491	1268	2236	4393	4396	1006	4399
18	1006	1384	71	4345	4353	4521	4345
19	1335	428	2224	4347	4385	1031	4373
20	1310	427	394	4373	4333	1335	4375
21	4333	1308	1477	4343	4374	2586	4336
22	4382	2910	1126	4375	491	1327	4343
23	4396	2487	428	4392	1310	4879	4347
24	4374	4121	1132	4366	1335	491	4377
25	4353	2954	2558	4377	1006	4225	2663

Table 6: Top 25 Nodes Ranked by Centrality Metric

Unfortunately, we lack specific information about what the nodes exactly represent (generator/substation/relay point...) or their geographical coordinates. This information could allow for contrast assumptions about node importance, (if they actually are central points with notable importance within the grid) or if they are geographically centered within the network. However, the discussions above

are still valid but this could be a limitation and focus point for future study.

Communities

The following figure shows the communities detected by the Louvain algorithm, ordered by community size (number of nodes present within the community).

The community ID (y-axis) is just a general ID that the algorithm assigns by default to each cluster of nodes. Again we are missing the true information of the exact nodes, therefore assumptions must be made to interpret the results that cannot be compared with actual locations or true node labels.

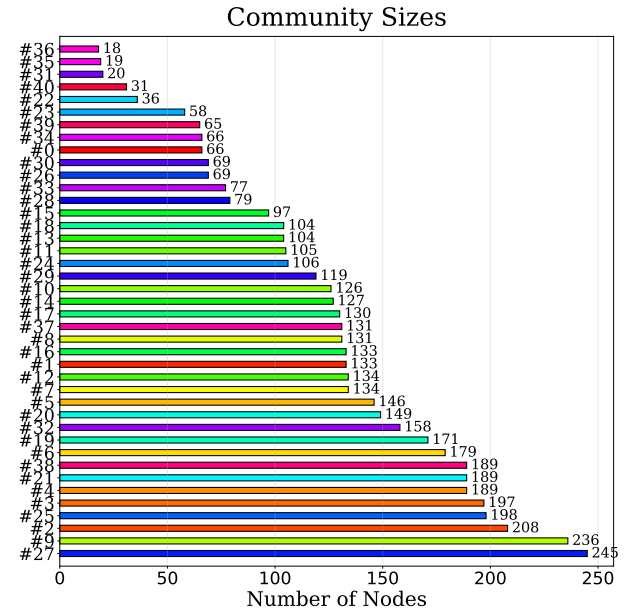


Figure 8: Community legend including the size (number of nodes) of each community.

Null Models

The following figures and numerical results complement the main analysis. These include the plots for the PDFs, CCDFs and mean overlap matrices for both ER and BA models. The values were averaged over the 10 realizations and the following figures show the outcomes.

The full description and analysis of these figures is included in Section IX NULL MODELS, this section only holds the figures.

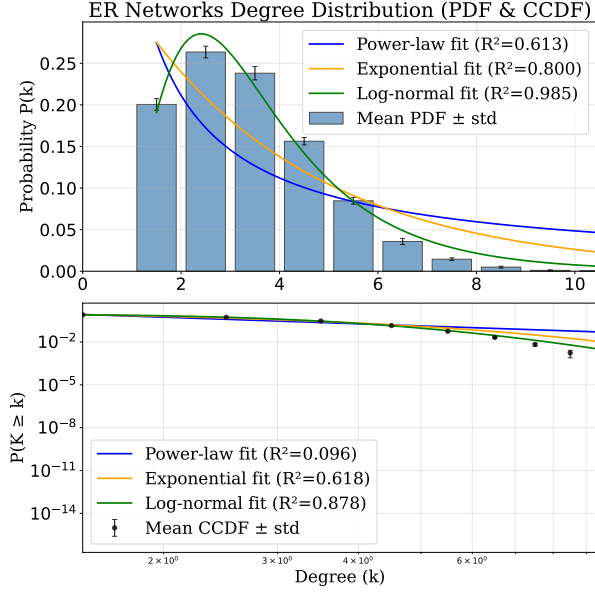


Figure 9: Mean PDF and CCDF for the simulated ER networks.

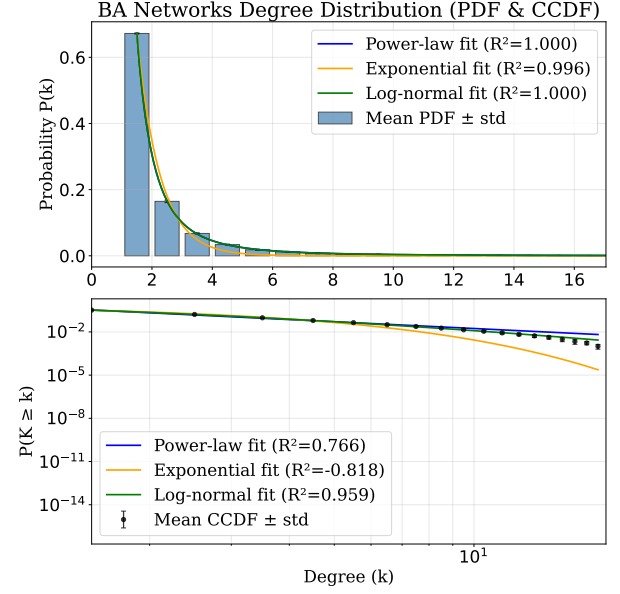


Figure 10: Mean PDF and CCDF for the simulated BA networks.

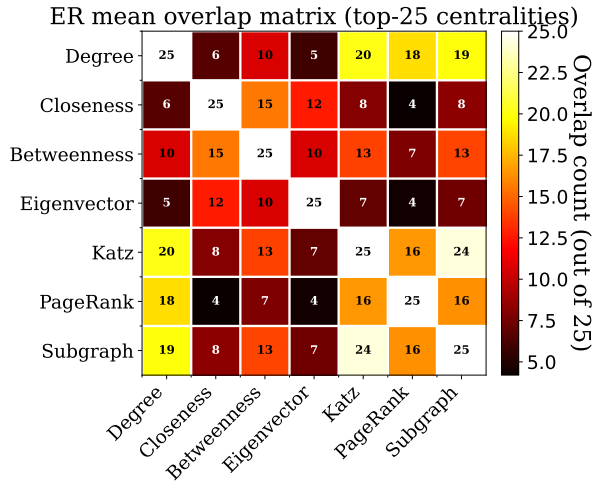


Figure 11: Mean overlap matrix for the top 25 nodes of ER networks.

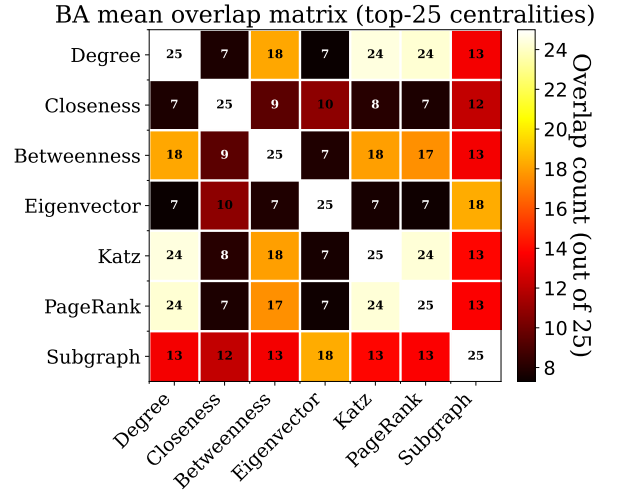


Figure 12: Mean overlap matrix for the top 25 nodes of BA networks.